

Supporting Information

Geography and life history mediate Pacific salmon responses to ocean warming and competition at sea

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1 Supplementary methods and results

1.1 Salmon data

We compiled data for pink, chum and sockeye salmon from a large public database of Pacific salmon spawner and recruitment time series (Greenberg (in prep)). These populations spanned 46.37 to 66.46 degrees in latitude and from 1948 to 2024. We discarded observations prior to 1950 due to concerns about data quality, and did not include time series that did not span at least brood years 1984-2014 and with minimal missing observation (i.e., > 80% of all years) (Figures S1-S3). We further excluded time series with known substantial hatchery influence to restrict our analysis to wild salmon populations. The resulting dataset included 66 sockeye populations, 28 chum salmon populations, and 33 pink salmon populations, grouped into three (chum) or four (sockeye and pink) large marine ecosystems/regions (Table S1). The most populations in a single region were West Coast sockeye (n=26) and the fewest were in the Bering Sea for pink salmon (n=4).

Unlike for other systems and species, estimates of spawner abundance for Fraser River were “effective female spawners” instead of total spawners, which accounts for the at times substantial pre-spawn mortality (i.e., mortality after arrival on the spawning grounds, but before spawning) and variable egg retention in this system. Additionally, return abundance estimates for Fraser River sockeye salmon account for en route mortality (i.e., mortality after freshwater entry, but before arrival on the spawning grounds), which can be considerable in certain years. While there is likely pre-spawn and en route mortality in other systems and species it is only systematically accounted for in the Fraser. Because of this, for modeling sockeye dynamics only, we allowed regional intrinsic productivity to arise from a common normal distribution that was allowed to differ from populations in other regions. In addition, for Fraser sockeye any confounding of freshwater temperature effects on return migration on estimates of marine temperature effects (e.g., in instances where they are correlated) in our analyses are accounted for by having adult recruitment explicitly account for freshwater migration mortality.

Sockeye can rear in freshwater for a variable length of time before migrating to sea, and both sockeye and chum can mature and return to spawn at variable ages. Where possible we used brood year specific estimates of age at ocean entry and return; when this data was not available we used estimates from adjacent years, or

the mean across all years if only a few years had brood year specific estimates. In some instances for chum, there were no age data available, and so we assumed a fixed four-year life cycle, following Malick et al. (2017) who found that assuming a fixed four year life cycle for chum introduced minimal bias into reconstructions of their population dynamics.

1.2 Sensitivity Analyses

Our analyses made a number of simplifying assumptions about processes that are hypothesized to influence Pacific salmon dynamics. Our intent was to quantify the statistical evidence for a relatively narrow set of hypotheses with strong a-priori justification (i.e., the model/hypotheses in the main text), and to then conduct sensitivity analyses to understand the extent to which our findings are sensitive to assumptions made during the analysis.

1.2.1 Alternative climate indices

Our analyses focused on coastal SST during early marine life as an index of ecological conditions during early marine life for each salmon population. This index has been repeatedly shown to correlate with pink, chum, and sockeye productivity across our study area Connors et al. (2020). However, previous analyses have also shown that the NPGO, that indexes sea surface height and is associated with primary production (e.g., nutrients, salinity, and chlorophyll-a.), can be associated with salmon productivity (Malick et al. 2017). To quantify the degree to which our inference about SST and pink salmon effects were confounded by other ocean basin scale processes (e.g., like those indexed by the NPGO) we repeated our analysis with the addition of the NPGO index. Specifically, we derived indices of the NPGO by averaging monthly values over the December-March period immediately prior to ocean entry, as in Mueter, Ware, and Peterman (2002) and Malick et al. (2017) and then included it as an additional covariate in either the stationary or “era” models. As with the SST index for sockeye the NPGO index was weighted by the proportion of each brood year cohort that outmigrated in each year. There was no strong positive correlation between the NPGO index and the SST or competitor covariates when all years were considered (-0.25 and 0.04 respectively). In contrast, our SST indices were moderately strongly correlated with another commonly considered ocean basin scale index, the PDO (0.4 across all years; 0.77 in the most recent time period), and so we did not consider it further in our analyses.

Stationary models that included the additional covariate indicated that the NPGO had a relatively small but typically positive relationship with productivity (Figures S7-S9), with the exception of Gulf of Alaska pink and chum where there was no evidence of a relationship (Figure S8 and S8), and Bering Sea chum where there was evidence of an inverse relationship. Importantly, considering the NPGO covariate had little effect on estimates of the effects of the SST and salmon competitor covariates. In the era models, there were larger magnitudes of positive NPGO effects in the most recent time period for sockeye (Figure S7), chum from more northern latitudes (Figure S8), and a negative effect for West Coast pink salmon (Figure S9). Overall, the spatial and temporal patterns we describe in the main text were insensitive to the simultaneous consideration of this additional covariate.

1.2.2 Alternative salmon competitor indices

To evaluate the sensitivity of our inference to alternative indices of inter- and intra-specific competition we considered four alternative competitor indices, in addition to the index of total North Pacific wide pink, chum, and sockeye salmon abundance we considered in the main text. These alternatives included: (1) the combined biomass of all sockeye, chum, and pink salmon in the North Pacific (based on conversion of adult abundance to biomass used in Ruggerone and Irvine 2018), and the North Pacific wide abundance of just (2) pink, (3) chum, or (3) sockeye. Each represents a slightly different hypothesis about the nature of competition among salmon at sea, for example, whether competition occurs among all pink, chum, and sockeye salmon, or primarily with pink salmon, or whether biomass better indexes potential competitors than abundance. In addition, analyses that only considered an index of competitors that was the same species as the focal populations were effectively just examining evidence for intra-specific competition.

We quantified the sensitivity of our results to these alternative assumptions by substituting each alternative index for the total abundance of pink, chum and sockeye we considered as a competitor index into the stationary models described in eqn. 1-4 in the main text. The indices were calculated in the same way as the original competitor index. We found that in general, while there were subtle differences in the estimated magnitude of competitor effects (and their interaction with SST), overall when estimated effects were strong in our baseline analysis they remained of a similar magnitude and direction when considering biomass instead of abundance but were often somewhat weaker when considering just individual species competitor indices (Figures S10-S12).

1.2.3 Detrended salmon competitor indices

The pink, chum, and sockeye salmon abundance time series we considered as an index of competition among salmon at sea exhibits an increasing time trend, in addition to high frequency year-to-year variation (Figure 1b in main text; Figure S13a). As a result it is possible that other unaccounted for processes that exhibit strong positive trends over the same period (e.g., marine mammal populations, ocean heat content) may confound our interpretation of a pink salmon competition “effect”. While the weight of evidence for the effect of competition with especially pink salmon on other salmon species is much stronger (Ruggerone et al. 2023) than for these other potential ecosystem changes, as a sensitivity analysis we also conducted an analysis with the temporal signal in the salmon time series removed. To do this we fit a General Additive Model to the North Pacific pink, chum, and sockeye salmon time series (Figure S13b) and used the resulting residuals of the model fit as an alternative, time trend removed, competitor index (Figure S13b). We then refit the stationary model (equation 1 in Main Text) with this alternative index to each of the sockeye, chum, and pink salmon spawner-recruitment data set.

Consistent with previous analyses that have fit models of salmon productivity to detrended salmon competitor indices Connors et al. (2020) statistical evidence of an inverse relationship between sockeye productivity and pink, chum and sockeye salmon abundance remained, though as expected since long term low frequency variation in the competitor time series was removed, the magnitude of predicted percentage change in recruits-per-spawner was smaller (Figure S14).

2 Figures

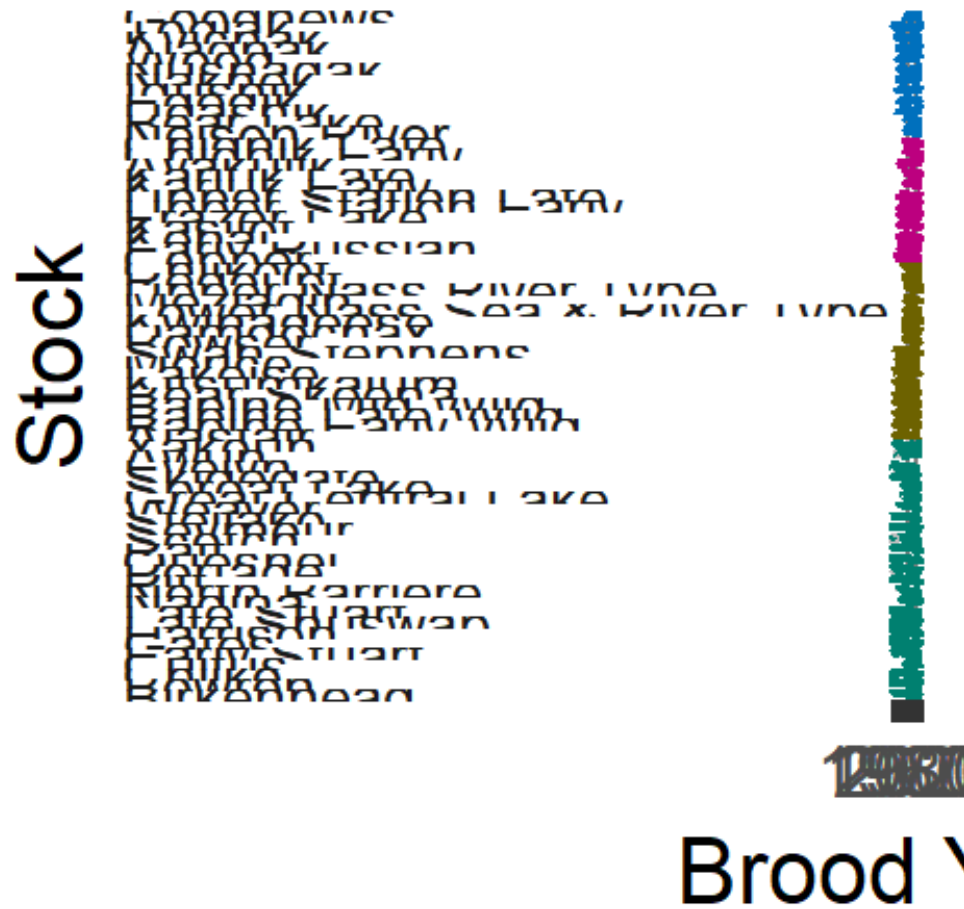


Figure S1: Sockeye salmon recruits-per-spawner timeseries used in the analysis. Missing brood years are in grey, regions are color coded (Table S1), and the temporal break point denoting the three “eras” considered are illustrated with vertical dashed lines.



Figure S3: Pink salmon recruits-per-spawner timeseries used in the analysis. Missing brood years are in grey, regions are color coded (Table S1), and the temporal break point denoting the three “eras” considered are illustrated with vertical dashed lines.

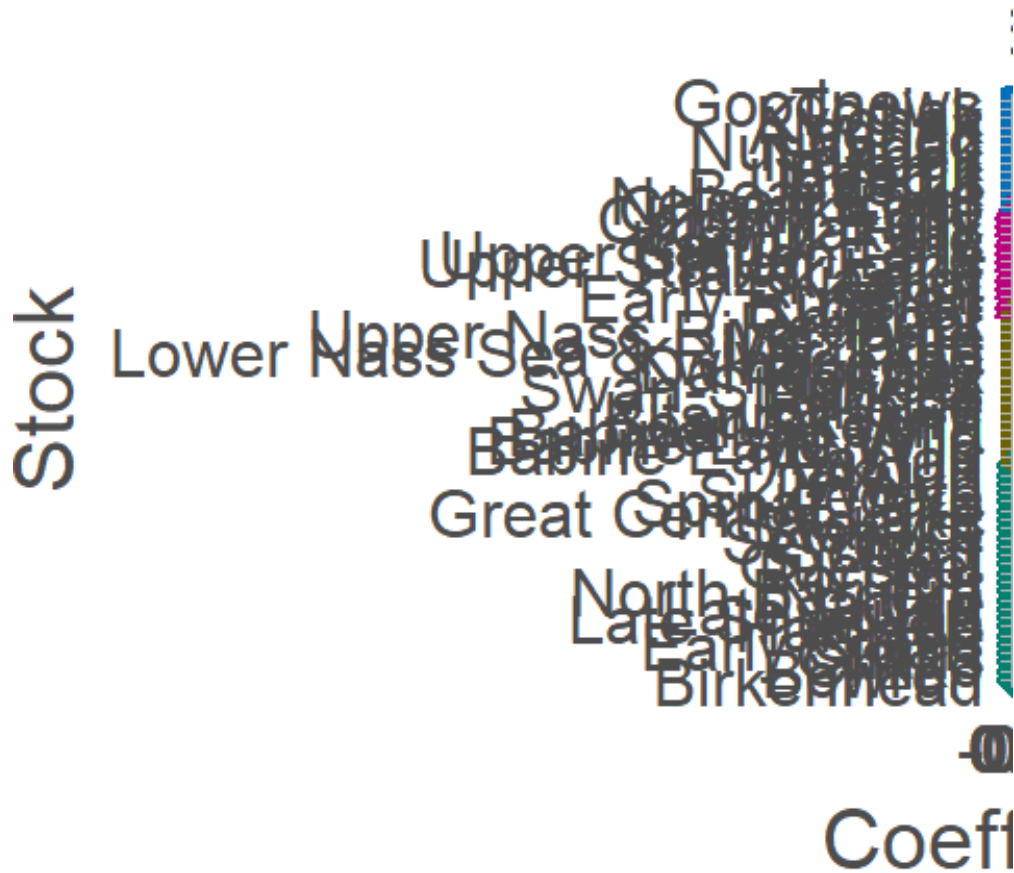


Figure S4: Sockeye posterior distribution summaries for SST and competition coefficients from the model described in the main text (equation 1). Filled points are posterior medians (points) and 95% credible intervals (horizontal lines) for the stock-specific effects of SST, and pink salmon competition, and open points are estimates from a simplified model that is fit individually and independently for each stock. Solid vertical lines give the posterior median for the ocean ecosystem level effects, which represent the mean effect across all stocks in an ocean ecosystem. Shaded box give the 95% credibility interval for the ecosystem-level effect with dashed lines indicating zero effect.

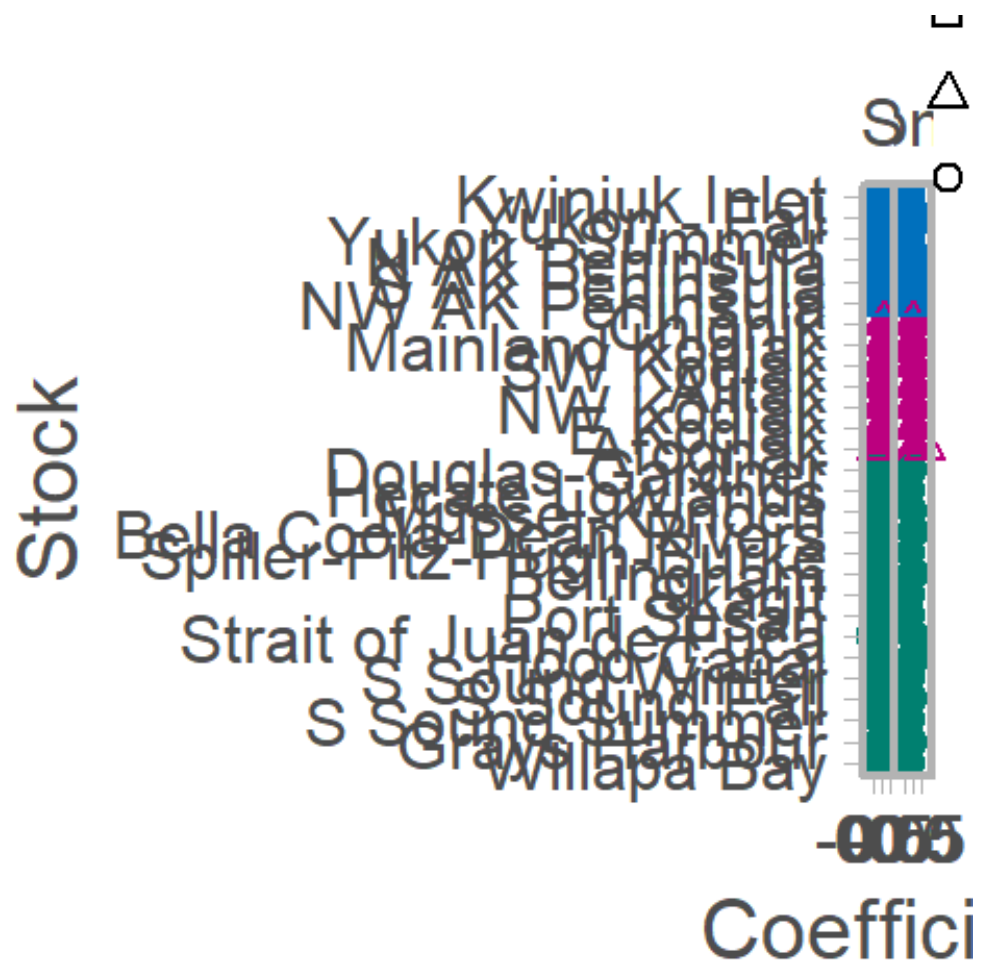


Figure S5: Same as Figure S4 but for chum salmon stocks.



Figure S6: Same as Figure S4 but for pink salmon stocks.

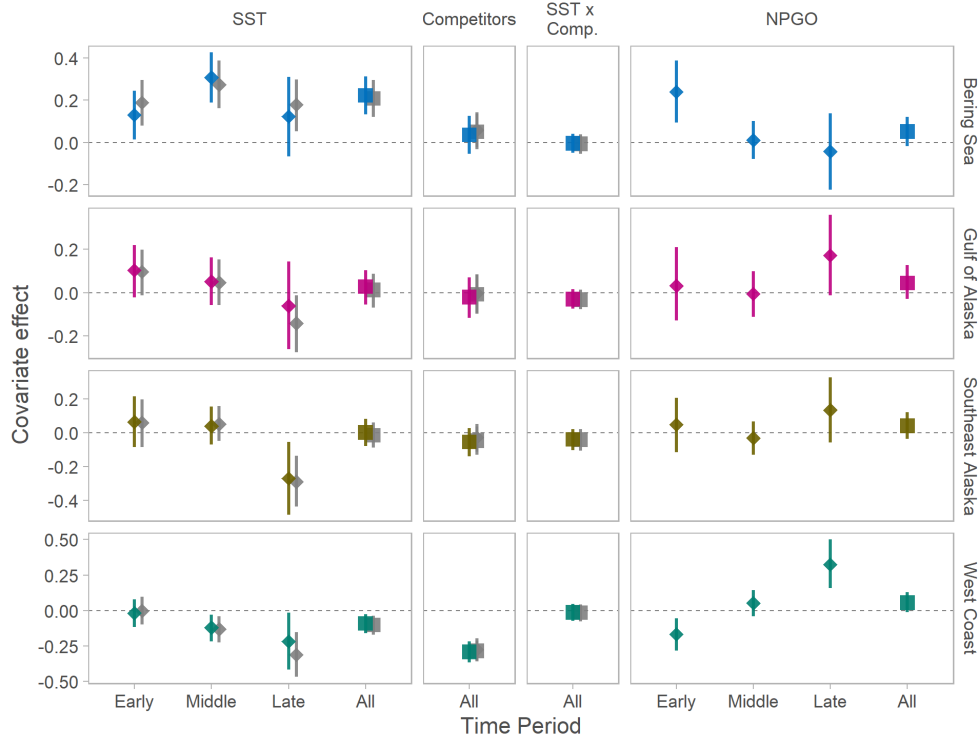


Figure S7: Posterior distribution summaries for covariate effects on sockeye salmon productivity when including the NPGO as an additional covariate in models (coloured points and lines) compared to original models with only SST, salmon competitors, and their interaction (grey points and lines). Points are medians of posterior distributions, lines are 95% credible intervals. Effect sizes estimated from time-varying models (pre-defined break points) are indicated with diamond-shaped points and labeled by time period: “Early” (1960-1988), “Middle” (1989-2010), or “Late” (2011-2016). Estimates from stationary models considering all years are indicated with square points and labeled “All”.

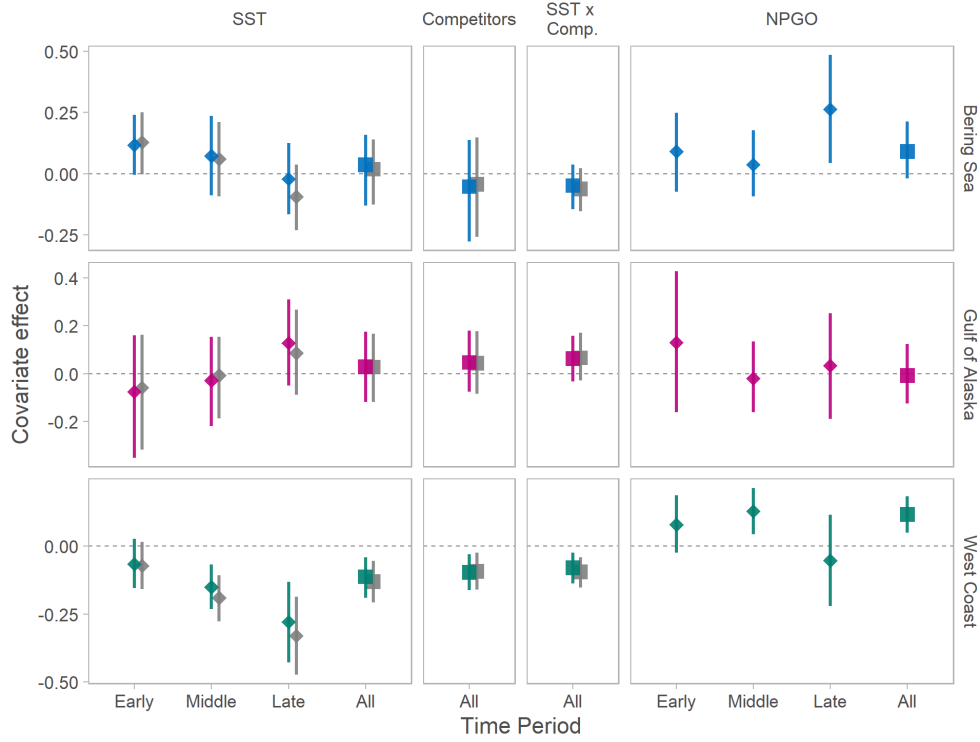


Figure S8: Same as Figure S7 but for chum salmon stocks.

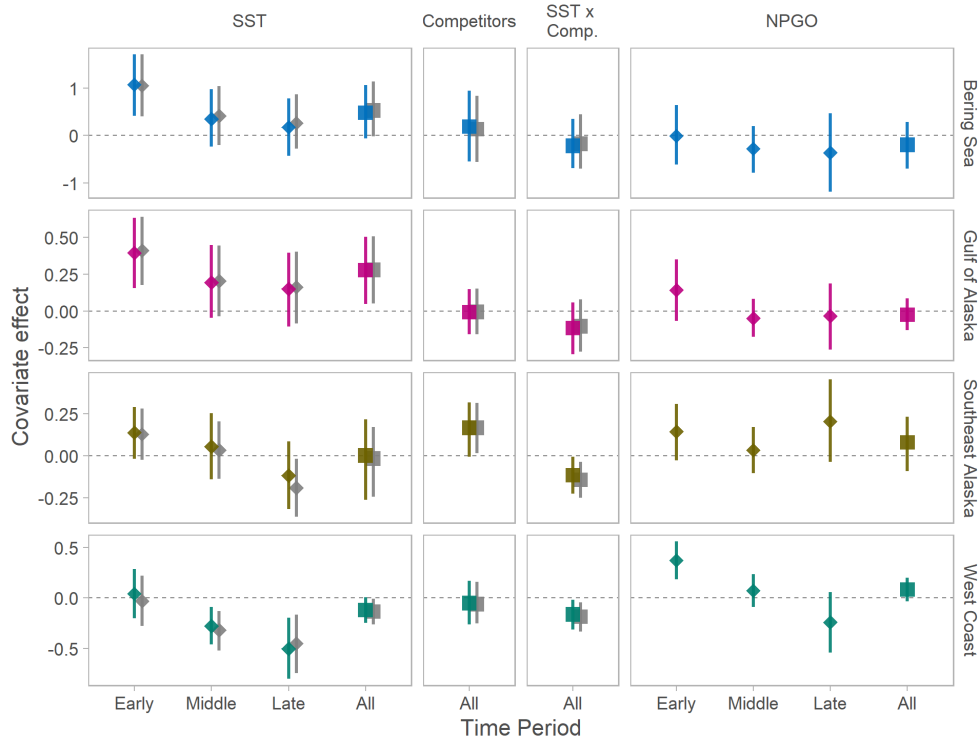


Figure S9: Same as Figure S7 but for pink salmon stocks.

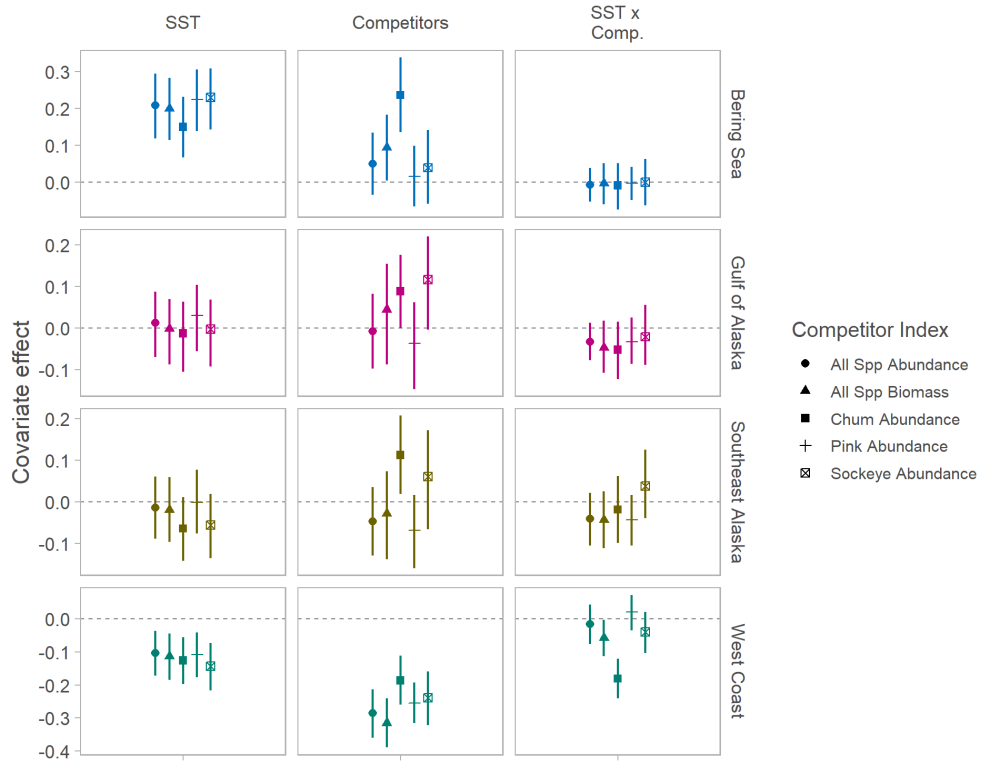


Figure S10: Posterior distribution summaries for covariate effects on sockeye salmon productivity across alternative indices of potential salmon competitors. Points are posterior means and vertical lines are 95% credible intervals.

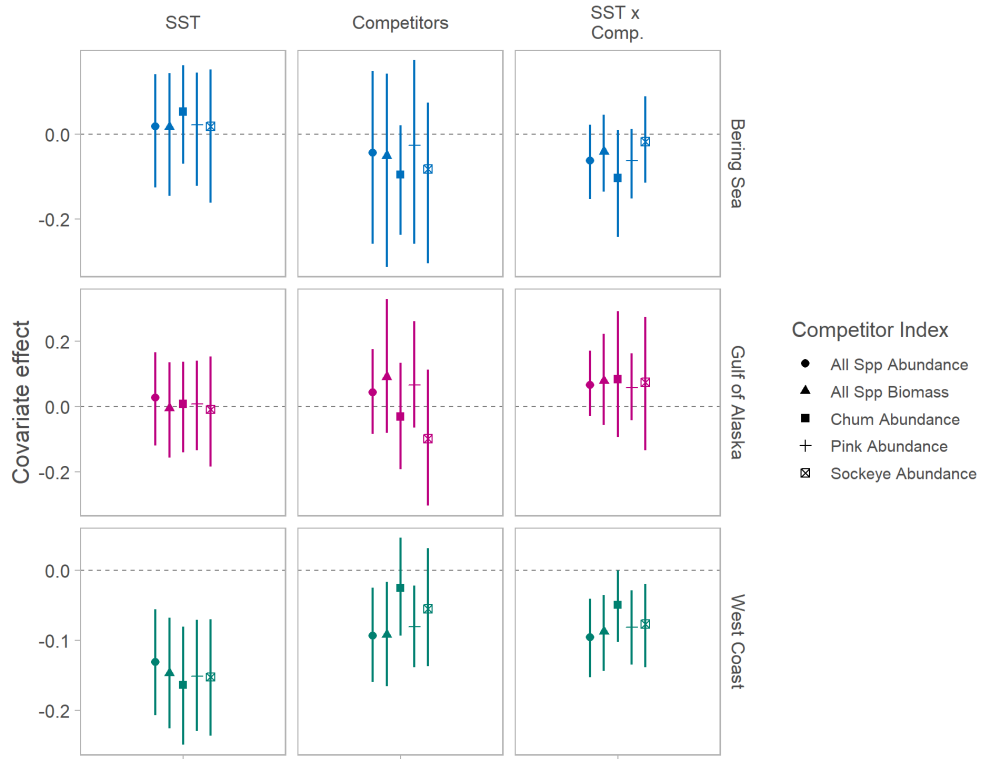


Figure S11: Posterior distribution summaries for covariate effects on chum salmon productivity across alternative indices of potential salmon competitors. Points are posterior means and vertical lines are 95% credible intervals.

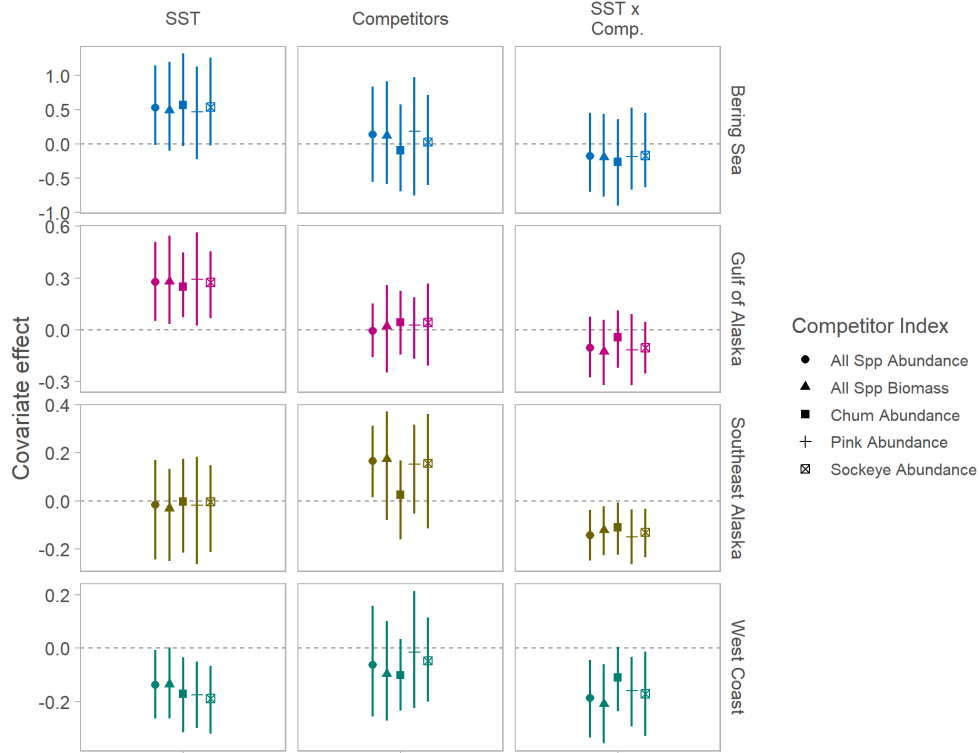


Figure S12: Posterior distribution summaries for covariate effects on pink salmon productivity across alternative indices of potential salmon competitors. Points are posterior means and vertical lines are 95% credible intervals.

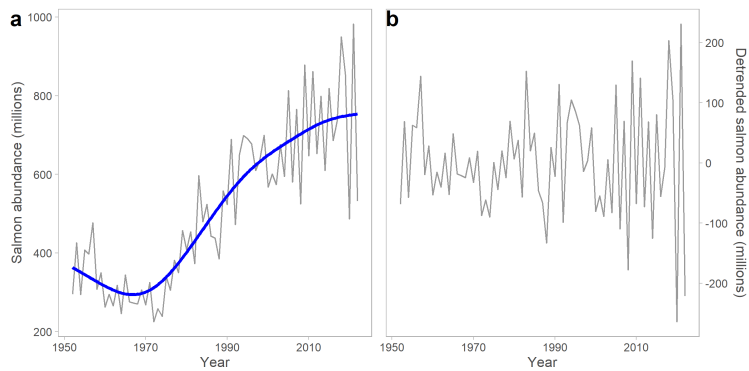


Figure S13: (a) North Pacific pink, chum, and sockeye salmon abundance over time (light grey) and GAM model fit to time series (blue) with resulting detrended competitor index (i.e., residuals of GAM model fit) show in (b).

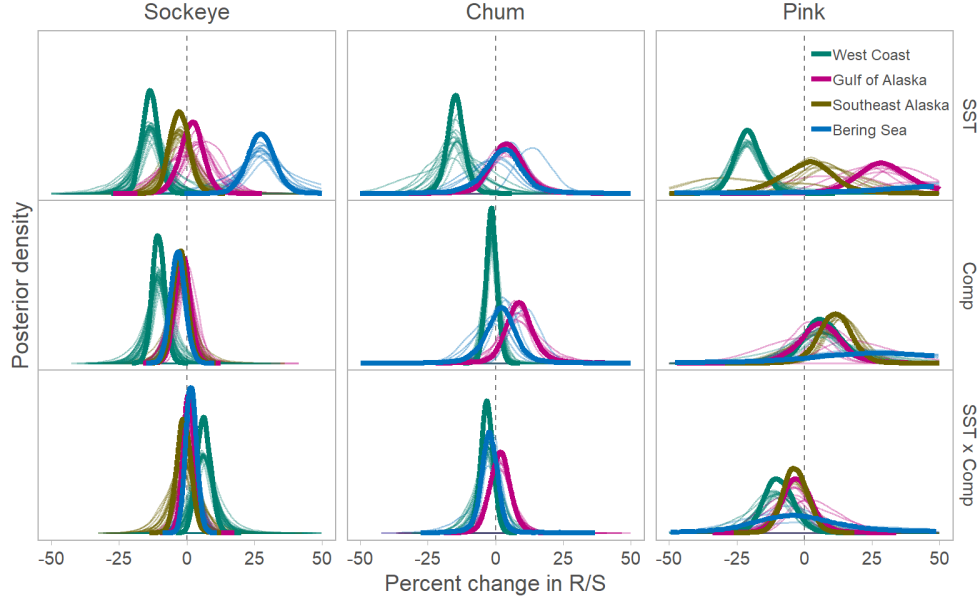


Figure S14: Effect of detrending salmon competitor index on estimates of impact of ocean warming and competition at sea. Posterior probability distributions of the predicted effect of SST and detrended competitors (Figure S13b) on sockeye, chum and pink salmon productivity (percentage change in adult recruits per spawner) by ocean ecosystem (same as Figure 2 in Main Text).

3 Tables

Table S1: Population considered in our analyses, ocean region, first and last brood year (BY), and approximate ocean entry point.

Species	Stock	Ocean Region	First BY	Last BY	Lat.	Lon.
Sockeye	Birkenhead	West Coast	1950	2015	49.12000	-123.0600
Sockeye	Bowron	West Coast	1950	2015	49.12000	-123.0600
Sockeye	Chilko	West Coast	1950	2015	49.12000	-123.0600
Sockeye	Cultus	West Coast	1950	2015	49.12000	-123.0600
Sockeye	Early Stuart	West Coast	1950	2015	49.12000	-123.0600
Sockeye	Gates	West Coast	1952	2015	49.12000	-123.0600
Sockeye	Harrison	West Coast	1950	2015	49.12000	-123.0600
Sockeye	Late Shuswap	West Coast	1950	2015	49.12000	-123.0600
Sockeye	Late Stuart	West Coast	1950	2015	49.12000	-123.0600
Sockeye	Nadina	West Coast	1973	2015	49.12000	-123.0600
Sockeye	North Barriere	West Coast	1958	2015	49.12000	-123.0600
Sockeye	Pitt	West Coast	1950	2015	49.12000	-123.0600
Sockeye	Portage	West Coast	1951	2015	49.12000	-123.0600
Sockeye	Quesnel	West Coast	1950	2015	49.12000	-123.0600
Sockeye	Raft	West Coast	1950	2015	49.12000	-123.0600
Sockeye	Scotch	West Coast	1952	2015	49.12000	-123.0600
Sockeye	Seymour	West Coast	1950	2015	49.12000	-123.0600
Sockeye	Stellako	West Coast	1950	2015	49.12000	-123.0600
Sockeye	Weaver	West Coast	1966	2015	49.12000	-123.0600
Sockeye	Great Central Lake	West Coast	1977	2015	49.24000	-124.8000

Table S1: Population considered in our analyses, ocean region, first and last brood year (BY), and approximate ocean entry point. (*continued*)

Species	Stock	Ocean Region	First BY	Last BY	Lat.	Lon.
Sockeye	Sproat Lake	West Coast	1977	2015	49.24000	-124.8000
Sockeye	Canoonna	West Coast	1975	2014	53.08000	-128.5550
Sockeye	Skidegate	West Coast	1956	2015	53.16700	-131.7890
Sockeye	Evelyn	West Coast	1954	2014	53.56000	-128.9580
Sockeye	Awun	West Coast	1957	2015	53.66500	-132.5230
Sockeye	Yakoun	West Coast	1957	2015	53.68200	-132.2350
Sockeye	Alastair	Southeast Alaska	1960	2015	54.22000	-129.8310
Sockeye	Babine Early Wild	Southeast Alaska	1960	2015	54.22000	-129.8310
Sockeye	Babine Late Wild	Southeast Alaska	1960	2015	54.22000	-129.8310
Sockeye	Babine Mid Wild	Southeast Alaska	1960	2015	54.22000	-129.8310
Sockeye	Bear-Skeena	Southeast Alaska	1960	2015	54.22000	-129.8310
Sockeye	Kitsumkalum	Southeast Alaska	1960	2015	54.22000	-129.8310
Sockeye	Lakelse	Southeast Alaska	1960	2015	54.22000	-129.8310
Sockeye	Morice	Southeast Alaska	1960	2015	54.22000	-129.8310
Sockeye	Swan-Stephens	Southeast Alaska	1960	2015	54.22000	-129.8310
Sockeye	Bowser	Southeast Alaska	1982	2015	54.99000	-130.0200
Sockeye	Damdochax	Southeast Alaska	1982	2015	54.99000	-130.0200
Sockeye	Kwinageese	Southeast Alaska	1982	2015	54.99000	-130.0200
Sockeye	Lower Nass Sea & River Type	Southeast Alaska	1982	2015	54.99000	-130.0200
Sockeye	Meziadin	Southeast Alaska	1982	2015	54.99000	-130.0200
Sockeye	Upper Nass River Type	Southeast Alaska	1982	2015	54.99000	-130.0200
Sockeye	Redoubt	Southeast Alaska	1982	2015	56.90000	-135.3300
Sockeye	Chilkoot	Southeast Alaska	1976	2015	58.92000	-135.2300
Sockeye	Copper	Gulf of Alaska	1961	2015	60.46000	-144.9600
Sockeye	Early Russian	Gulf of Alaska	1965	2015	60.54000	-151.2800
Sockeye	Kenai	Gulf of Alaska	1968	2015	60.54000	-151.2800
Sockeye	Kasilof	Gulf of Alaska	1968	2015	60.39000	-151.3000
Sockeye	Upper Station Early	Gulf of Alaska	1969	2015	57.06000	-154.3600
Sockeye	Upper Station Late	Gulf of Alaska	1970	2015	57.06000	-154.3600
Sockeye	Karluk Early	Gulf of Alaska	1982	2015	57.57000	-154.4500
Sockeye	Karluk Late	Gulf of Alaska	1982	2015	57.57000	-154.4500
Sockeye	Ayakulik	Gulf of Alaska	1966	2015	57.20000	-154.5400
Sockeye	Chignik Early	Gulf of Alaska	1983	2015	56.28000	-158.6400
Sockeye	Chignik Late	Gulf of Alaska	1983	2015	56.28000	-158.6400
Sockeye	Nelson	Bering Sea	1982	2015	55.96000	-161.2800
Sockeye	Bear	Bering Sea	1982	2015	56.16000	-160.4500
Sockeye	Kvichak	Bering Sea	1963	2014	57.50000	-157.6000
Sockeye	Alagnak	Bering Sea	1963	2014	58.25000	-157.3800
Sockeye	Wood	Bering Sea	1963	2014	58.68000	-158.8000
Sockeye	Naknek	Bering Sea	1963	2014	58.78000	-157.0800
Sockeye	Igushik	Bering Sea	1963	2014	58.82000	-158.6200
Sockeye	Nushagak	Bering Sea	1963	2014	58.82000	-158.4900
Sockeye	Egegik	Bering Sea	1963	2014	58.85000	-157.0300
Sockeye	Ugashik	Bering Sea	1963	2014	58.85000	-158.5000
Sockeye	Togiak	Bering Sea	1956	2015	58.95000	-160.4500

Table S1: Population considered in our analyses, ocean region, first and last brood year (BY), and approximate ocean entry point. (*continued*)

Species	Stock	Ocean Region	First BY	Last BY	Lat.	Lon.
Sockeye	Goodnews	Bering Sea	1981	2015	59.11000	-161.6200
Chum	Willapa Bay	West Coast	1968	2016	46.37330	-123.9575
Chum	Grays Harbour	West Coast	1969	2016	46.95083	-124.0497
Chum	S Sound Summer	West Coast	1968	2016	47.04200	-122.8350
Chum	S Sound Fall	West Coast	1968	2016	47.04200	-122.8350
Chum	S Sound Winter	West Coast	1968	2016	47.04200	-122.8350
Chum	Hood Canal	West Coast	1968	2016	47.56440	-122.9956
Chum	Strait of Juan de Fuca	West Coast	1968	2016	48.16200	-123.5900
Chum	Port Susan	West Coast	1968	2016	48.23861	-122.3758
Chum	Skagit	West Coast	1968	2016	48.38750	-122.3656
Chum	Bellingham	West Coast	1968	2016	48.77500	-122.5978
Chum	Spiller-Fitz-Hugh-Burke	West Coast	1954	2016	52.16000	-128.1400
Chum	Bella Coola-Dean Rivers	West Coast	1954	2016	52.38000	-126.7800
Chum	Mussel-Kynoch	West Coast	1954	2015	52.61000	-128.5200
Chum	Hecate Lowlands	West Coast	1954	2016	53.42000	-129.2500
Chum	Douglas-Gardner	West Coast	1954	2015	53.96000	-128.6800
Chum	Afognak	Gulf of Alaska	1984	2019	58.40000	-152.0000
Chum	E Kodiak	Gulf of Alaska	1984	2019	57.08986	-153.1879
Chum	NW Kodiak	Gulf of Alaska	1984	2019	57.85684	-153.8829
Chum	Alitak	Gulf of Alaska	1984	2019	56.88528	-154.1217
Chum	SW Kodiak	Gulf of Alaska	1984	2019	57.26451	-154.6132
Chum	Mainland Kodiak	Gulf of Alaska	1979	2019	58.03333	-154.6644
Chum	Chignik	Gulf of Alaska	1980	2019	56.59639	-157.9933
Chum	NW AK Peninsula	Bering Sea	1962	2019	54.98200	-163.3700
Chum	S AK Peninsula	Bering Sea	1962	2019	55.50000	-161.0000
Chum	N AK Peninsula	Bering Sea	1962	2019	56.46000	-159.6800
Chum	Yukon - Summer	Bering Sea	1978	2017	62.55000	-165.0000
Chum	Yukon - Fall	Bering Sea	1974	2019	62.55000	-165.0000
Chum	Kwiniuk Inlet	Bering Sea	1965	2018	64.72000	-161.9600
Pink	Nisqually-Odd	West Coast	1959	2019	47.10100	-122.7060
Pink	S South Misc.-Odd	West Coast	1959	2019	47.16000	-122.9100
Pink	Puyallup-Odd	West Coast	1959	2019	47.26972	-122.4272
Pink	Hood Canal-Odd	West Coast	1959	2019	47.56440	-122.9956
Pink	Snohomish-Odd	West Coast	1959	2019	48.02139	-122.2072
Pink	Dungeness-Odd	West Coast	1959	2019	48.15080	-123.1333
Pink	Stillaguamish-Odd	West Coast	1959	2019	48.23861	-122.3758
Pink	Skagit-Odd	West Coast	1959	2019	48.38750	-122.3656
Pink	Nooksack-Odd	West Coast	1959	2019	48.77500	-122.5978
Pink	Fraser-Odd	West Coast	1961	2019	49.12333	-123.0569
Pink	Hecate Lowlands-Even	West Coast	1954	2018	52.45000	-129.1000
Pink	Hecate Strait-Fjords-Even	West Coast	1954	2018	52.45000	-129.1000
Pink	N Haida Gwaii-Even	Southeast Alaska	1954	2018	54.09000	-132.1400
Pink	S Southeast Alaska-Even	Southeast Alaska	1960	2020	55.71556	-131.8714
Pink	S Southeast Alaska-Odd	Southeast Alaska	1961	2019	55.71556	-131.8714
Pink	N Southeast Alaska Outside-Even	Southeast Alaska	1960	2020	57.17694	-135.7869

Table S1: Population considered in our analyses, ocean region, first and last brood year (BY), and approximate ocean entry point. (*continued*)

Species	Stock	Ocean Region	First BY	Last BY	Lat.	Lon.
Pink	N Southeast Alaska Outside-Odd	Southeast Alaska	1961	2019	57.17694	-135.7869
Pink	N Southeast Alaska Inside-Even	Southeast Alaska	1960	2020	57.57528	-134.7503
Pink	N Southeast Alaska Inside-Odd	Southeast Alaska	1961	2019	57.57528	-134.7503
Pink	Auke Creek-Even	Southeast Alaska	1978	2016	58.37600	-134.6500
Pink	Auke Creek-Odd	Southeast Alaska	1979	2015	58.37600	-134.6500
Pink	Prince William Sound-Even	Gulf of Alaska	1960	2016	60.50000	-147.0000
Pink	Prince William Sound-Odd	Gulf of Alaska	1961	2017	60.50000	-147.0000
Pink	Kodiak-Even	Gulf of Alaska	1978	2020	58.03333	-154.6644
Pink	Kodiak-Odd	Gulf of Alaska	1979	2019	58.03333	-154.6644
Pink	Chignik-Even	Gulf of Alaska	1980	2020	56.38528	-158.2261
Pink	Chignik-Odd	Gulf of Alaska	1981	2019	56.38528	-158.2261
Pink	S AKPen-Even	Gulf of Alaska	1962	2020	55.00000	-161.0000
Pink	S AKPen-Odd	Gulf of Alaska	1963	2019	55.00000	-161.0000
Pink	NW AK Peninsula-Even	Bering Sea	1962	2020	54.98200	-163.3700
Pink	NW AK Peninsula-Odd	Bering Sea	1963	2019	54.98200	-163.3700
Pink	Kwiniuk-Even	Bering Sea	1966	2020	64.68000	-162.0100
Pink	Kwiniuk-Odd	Bering Sea	1965	2019	64.68000	-162.0100

Table S2: Estimated mean effect sizes by region for SST and Competitor covariates, and interaction term, from stationary models.

Species	Ocean Region	Coefficient	Lower CI	Mean	Upper CI	Mean % change in R/S
sockeye	West Coast	SST	-0.17	-0.10	-0.04	-9.76
sockeye	Southeast Alaska	SST	-0.09	-0.01	0.06	-1.43
sockeye	Gulf of Alaska	SST	-0.07	0.01	0.09	1.25
sockeye	Bering Sea	SST	0.12	0.21	0.29	23.21
sockeye	West Coast	Comp	-0.36	-0.29	-0.21	-24.82
sockeye	Southeast Alaska	Comp	-0.13	-0.05	0.03	-4.53
sockeye	Gulf of Alaska	Comp	-0.10	-0.01	0.08	-0.80
sockeye	Bering Sea	Comp	-0.03	0.05	0.13	5.10
sockeye	West Coast	SST x Comp	-0.08	-0.01	0.04	-1.46
sockeye	Southeast Alaska	SST x Comp	-0.10	-0.04	0.02	-4.03
sockeye	Gulf of Alaska	SST x Comp	-0.08	-0.03	0.01	-3.24
sockeye	Bering Sea	SST x Comp	-0.05	-0.01	0.04	-0.65
chum	West Coast	SST	-0.21	-0.13	-0.06	-12.29
chum	Gulf of Alaska	SST	-0.12	0.03	0.17	2.73
chum	Bering Sea	SST	-0.13	0.02	0.14	1.91
chum	West Coast	Comp	-0.16	-0.09	-0.02	-8.91
chum	Gulf of Alaska	Comp	-0.08	0.04	0.18	4.48
chum	Bering Sea	Comp	-0.26	-0.04	0.15	-4.22
chum	West Coast	SST x Comp	-0.15	-0.10	-0.04	-9.10
chum	Gulf of Alaska	SST x Comp	-0.03	0.07	0.17	6.76
chum	Bering Sea	SST x Comp	-0.15	-0.06	0.02	-5.99

Table S2: Estimated mean effect sizes by region for SST and Competitor covariates, and interaction term, from stationary models. (*continued*)

Species	Ocean Region	Coefficient	Lower CI	Mean	Upper CI	Mean % change in R/S
pink	West Coast	SST	-0.26	-0.14	-0.01	-12.76
pink	Southeast Alaska	SST	-0.24	-0.02	0.17	-1.64
pink	Gulf of Alaska	SST	0.05	0.28	0.51	32.11
pink	Bering Sea	SST	-0.02	0.52	1.14	69.04
pink	West Coast	Comp	-0.25	-0.06	0.16	-5.94
pink	Southeast Alaska	Comp	0.02	0.17	0.31	18.15
pink	Gulf of Alaska	Comp	-0.16	-0.01	0.15	-0.60
pink	Bering Sea	Comp	-0.56	0.13	0.84	14.35
pink	West Coast	SST x Comp	-0.33	-0.19	-0.04	-16.99
pink	Southeast Alaska	SST x Comp	-0.25	-0.14	-0.04	-13.30
pink	Gulf of Alaska	SST x Comp	-0.27	-0.10	0.08	-9.89
pink	Bering Sea	SST x Comp	-0.71	-0.18	0.45	-16.15

Table S3: Estimated stock-specific mean effect sizes for SST and Competitor covariates, and interaction term, from stationary models.

Species	Stock	Ocean Region	Coefficient	Lower CI	Mean	Upper CI	Mean % change in R/S
sockeye	Birkenhead	West Coast	SST	-0.26	-0.11	0.01	-10.85
sockeye	Bowron	West Coast	SST	-0.24	-0.10	0.04	-9.86
sockeye	Chilko	West Coast	SST	-0.24	-0.10	0.00	-9.94
sockeye	Cultus	West Coast	SST	-0.40	-0.15	-0.02	-14.03
sockeye	Early Stuart	West Coast	SST	-0.25	-0.12	-0.02	-11.33
sockeye	Gates	West Coast	SST	-0.29	-0.12	0.00	-10.90
sockeye	Harrison	West Coast	SST	-0.23	-0.08	0.08	-7.91
sockeye	Late Shuswap	West Coast	SST	-0.33	-0.14	-0.02	-12.73
sockeye	Late Stuart	West Coast	SST	-0.22	-0.09	0.07	-8.70
sockeye	Nadina	West Coast	SST	-0.20	-0.08	0.08	-7.81
sockeye	North Barriere	West Coast	SST	-0.27	-0.12	0.01	-11.65
sockeye	Pitt	West Coast	SST	-0.18	-0.08	0.07	-7.55
sockeye	Portage	West Coast	SST	-0.28	-0.12	0.01	-11.41
sockeye	Quesnel	West Coast	SST	-0.27	-0.12	0.00	-11.35
sockeye	Raft	West Coast	SST	-0.22	-0.10	0.03	-9.57
sockeye	Scotch	West Coast	SST	-0.30	-0.13	-0.01	-11.76
sockeye	Seymour	West Coast	SST	-0.26	-0.11	0.01	-10.82
sockeye	Stellako	West Coast	SST	-0.25	-0.12	-0.01	-11.32
sockeye	Weaver	West Coast	SST	-0.27	-0.11	0.04	-10.28
sockeye	Great Central Lake	West Coast	SST	-0.26	-0.09	0.05	-8.69
sockeye	Sproat Lake	West Coast	SST	-0.25	-0.10	0.03	-9.28
sockeye	Skidegate	West Coast	SST	-0.21	-0.09	0.02	-8.92
sockeye	Evelyn	West Coast	SST	-0.29	-0.12	0.01	-11.03
sockeye	Awun	West Coast	SST	-0.14	0.00	0.27	0.46
sockeye	Yakoun	West Coast	SST	-0.16	-0.06	0.11	-5.73

Table S3: Estimated stock-specific mean effect sizes for SST and Competitor covariates, and interaction term, from stationary models. *(continued)*

Species	Stock	Ocean Region	Coefficient	Lower CI	Mean	Upper CI	Mean % change in R/S
sockeye	Alastair	Southeast Alaska	SST	-0.13	-0.02	0.10	-1.57
sockeye	Babine Early Wild	Southeast Alaska	SST	-0.19	-0.05	0.06	-4.59
sockeye	Babine Late Wild	Southeast Alaska	SST	-0.21	-0.05	0.07	-5.00
sockeye	Babine Mid Wild	Southeast Alaska	SST	-0.20	-0.05	0.07	-4.62
sockeye	Bear-Skeena	Southeast Alaska	SST	-0.10	0.01	0.13	0.91
sockeye	Kitsumkalum	Southeast Alaska	SST	-0.07	0.02	0.13	1.83
sockeye	Lakelse	Southeast Alaska	SST	-0.12	-0.01	0.09	-1.39
sockeye	Morice	Southeast Alaska	SST	-0.07	0.04	0.22	3.79
sockeye	Swan-Stephens	Southeast Alaska	SST	-0.12	-0.01	0.10	-1.24
sockeye	Bowser	Southeast Alaska	SST	-0.18	-0.04	0.06	-4.19
sockeye	Damdochax	Southeast Alaska	SST	-0.14	-0.03	0.07	-2.87
sockeye	Kwinageese	Southeast Alaska	SST	-0.14	-0.01	0.13	-1.14
sockeye	Lower Nass Sea & River Type	Southeast Alaska	SST	-0.10	0.02	0.18	1.61
sockeye	Meziadin	Southeast Alaska	SST	-0.14	-0.02	0.07	-2.27
sockeye	Upper Nass River Type	Southeast Alaska	SST	-0.14	-0.02	0.09	-2.03
sockeye	Redoubt	Southeast Alaska	SST	-0.15	-0.02	0.11	-2.06
sockeye	Chilkoot	Southeast Alaska	SST	-0.10	0.01	0.14	0.84
sockeye	Copper	Gulf of Alaska	SST	0.00	0.08	0.18	8.64
sockeye	Early Russian	Gulf of Alaska	SST	-0.08	0.04	0.15	3.60
sockeye	Kenai	Gulf of Alaska	SST	-0.17	-0.03	0.07	-3.44
sockeye	Kasilof	Gulf of Alaska	SST	-0.04	0.04	0.13	4.46
sockeye	Frazer Lake	Gulf of Alaska	SST	-0.20	-0.02	0.10	-1.72
sockeye	Upper Station Early	Gulf of Alaska	SST	-0.06	0.04	0.16	4.06
sockeye	Upper Station Late	Gulf of Alaska	SST	-0.10	0.01	0.11	0.63
sockeye	Karluk Early	Gulf of Alaska	SST	-0.14	-0.01	0.09	-0.74
sockeye	Karluk Late	Gulf of Alaska	SST	-0.09	0.02	0.12	1.68
sockeye	Ayakulik	Gulf of Alaska	SST	-0.06	0.04	0.15	4.47
sockeye	Chignik Early	Gulf of Alaska	SST	-0.19	-0.03	0.08	-2.71
sockeye	Chignik Late	Gulf of Alaska	SST	-0.20	-0.03	0.08	-3.30
sockeye	Nelson River	Bering Sea	SST	-0.06	0.13	0.27	14.14
sockeye	Bear Lake	Bering Sea	SST	0.01	0.18	0.30	19.60
sockeye	Ugashik	Bering Sea	SST	0.17	0.30	0.50	35.29
sockeye	Egegik	Bering Sea	SST	0.07	0.21	0.34	23.35
sockeye	Igushik	Bering Sea	SST	0.07	0.20	0.32	22.17
sockeye	Naknek	Bering Sea	SST	0.10	0.21	0.32	23.48
sockeye	Nushagak	Bering Sea	SST	0.10	0.22	0.37	24.82
sockeye	Wood	Bering Sea	SST	0.15	0.24	0.35	27.18
sockeye	Alagnak	Bering Sea	SST	0.13	0.25	0.41	28.57
sockeye	Kvichak	Bering Sea	SST	-0.02	0.15	0.28	16.77
sockeye	Togiak	Bering Sea	SST	0.10	0.20	0.30	22.54
sockeye	Goodnews	Bering Sea	SST	0.05	0.20	0.35	22.63

Table S3: Estimated stock-specific mean effect sizes for SST and Competitor covariates, and interaction term, from stationary models. *(continued)*

Species	Stock	Ocean Region	Coefficient	Lower CI	Mean	Upper CI	Mean % change in R/S
sockeye	Birkenhead	West Coast	Comp	-0.49	-0.31	-0.18	-26.58
sockeye	Bowron	West Coast	Comp	-0.46	-0.30	-0.16	-25.55
sockeye	Chilko	West Coast	Comp	-0.39	-0.27	-0.12	-23.81
sockeye	Cultus	West Coast	Comp	-0.49	-0.31	-0.15	-26.69
sockeye	Early Stuart	West Coast	Comp	-0.43	-0.30	-0.19	-25.82
sockeye	Gates	West Coast	Comp	-0.42	-0.28	-0.11	-24.33
sockeye	Harrison	West Coast	Comp	-0.38	-0.25	-0.02	-21.77
sockeye	Late Shuswap	West Coast	Comp	-0.48	-0.30	-0.15	-26.23
sockeye	Late Stuart	West Coast	Comp	-0.46	-0.29	-0.15	-25.13
sockeye	Nadina	West Coast	Comp	-0.41	-0.27	-0.13	-23.97
sockeye	North Barriere	West Coast	Comp	-0.55	-0.33	-0.21	-28.21
sockeye	Pitt	West Coast	Comp	-0.41	-0.28	-0.16	-24.46
sockeye	Portage	West Coast	Comp	-0.53	-0.32	-0.18	-27.16
sockeye	Quesnel	West Coast	Comp	-0.50	-0.32	-0.20	-27.26
sockeye	Raft	West Coast	Comp	-0.44	-0.29	-0.17	-25.44
sockeye	Scotch	West Coast	Comp	-0.46	-0.30	-0.16	-25.96
sockeye	Seymour	West Coast	Comp	-0.46	-0.30	-0.16	-25.74
sockeye	Stellako	West Coast	Comp	-0.48	-0.31	-0.20	-27.01
sockeye	Weaver	West Coast	Comp	-0.46	-0.30	-0.15	-25.56
sockeye	Great Central Lake	West Coast	Comp	-0.36	-0.23	0.00	-20.82
sockeye	Sproat Lake	West Coast	Comp	-0.38	-0.26	-0.08	-22.82
sockeye	Skidegate	West Coast	Comp	-0.39	-0.27	-0.11	-23.32
sockeye	Evelyn	West Coast	Comp	-0.45	-0.29	-0.14	-25.51
sockeye	Awun	West Coast	Comp	-0.33	-0.18	0.09	-16.24
sockeye	Yakoun	West Coast	Comp	-0.41	-0.27	-0.15	-23.79
sockeye	Alastair	Southeast Alaska	Comp	-0.18	-0.04	0.10	-4.17
sockeye	Babine Early Wild	Southeast Alaska	Comp	-0.31	-0.10	0.03	-9.57
sockeye	Babine Late Wild	Southeast Alaska	Comp	-0.33	-0.11	0.03	-10.25
sockeye	Babine Mid Wild	Southeast Alaska	Comp	-0.30	-0.10	0.04	-9.12
sockeye	Bear-Skeena	Southeast Alaska	Comp	-0.12	0.00	0.17	-0.16
sockeye	Kitsumkalum	Southeast Alaska	Comp	-0.11	0.01	0.16	0.61
sockeye	Lakelse	Southeast Alaska	Comp	-0.17	-0.04	0.08	-4.15
sockeye	Morice	Southeast Alaska	Comp	-0.12	0.01	0.19	1.12
sockeye	Swan-Stephens	Southeast Alaska	Comp	-0.19	-0.05	0.08	-4.65
sockeye	Bowser	Southeast Alaska	Comp	-0.23	-0.08	0.04	-7.58
sockeye	Damdochax	Southeast Alaska	Comp	-0.22	-0.08	0.04	-7.27
sockeye	Kwinageese	Southeast Alaska	Comp	-0.21	-0.04	0.12	-4.23
sockeye	Lower Nass Sea & River Type	Southeast Alaska	Comp	-0.15	0.00	0.19	-0.47
sockeye	Meziadin	Southeast Alaska	Comp	-0.16	-0.05	0.07	-4.44
sockeye	Upper Nass River Type	Southeast Alaska	Comp	-0.19	-0.05	0.08	-5.19
sockeye	Redoubt	Southeast Alaska	Comp	-0.23	-0.06	0.09	-5.68
sockeye	Chilkoot	Southeast Alaska	Comp	-0.13	-0.01	0.16	-0.93

Table S3: Estimated stock-specific mean effect sizes for SST and Competitor covariates, and interaction term, from stationary models. *(continued)*

Species	Stock	Ocean Region	Coefficient	Lower CI	Mean	Upper CI	Mean % change in R/S
sockeye	Copper	Gulf of Alaska	Comp	-0.03	0.09	0.23	8.89
sockeye	Early Russian	Gulf of Alaska	Comp	-0.11	0.02	0.18	2.39
sockeye	Kenai	Gulf of Alaska	Comp	-0.24	-0.08	0.04	-7.36
sockeye	Kasilof	Gulf of Alaska	Comp	-0.06	0.03	0.15	3.55
sockeye	Frazer Lake	Gulf of Alaska	Comp	-0.20	-0.03	0.13	-3.17
sockeye	Upper Station	Gulf of Alaska	Comp	-0.10	0.02	0.16	2.24
sockeye	Early Upper Station Late	Gulf of Alaska	Comp	-0.17	-0.03	0.09	-3.06
sockeye	Karluk Early	Gulf of Alaska	Comp	-0.18	-0.04	0.08	-3.84
sockeye	Karluk Late	Gulf of Alaska	Comp	-0.13	-0.01	0.10	-1.06
sockeye	Ayakulik	Gulf of Alaska	Comp	-0.09	0.03	0.19	3.51
sockeye	Chignik Early	Gulf of Alaska	Comp	-0.22	-0.06	0.06	-5.85
sockeye	Chignik Late	Gulf of Alaska	Comp	-0.19	-0.05	0.06	-5.29
sockeye	Nelson River	Bering Sea	Comp	-0.16	0.00	0.11	-0.37
sockeye	Bear Lake	Bering Sea	Comp	-0.07	0.04	0.14	3.72
sockeye	Ugashik	Bering Sea	Comp	0.00	0.11	0.30	12.08
sockeye	Egegik	Bering Sea	Comp	-0.05	0.06	0.18	5.73
sockeye	Igushik	Bering Sea	Comp	-0.07	0.04	0.14	4.20
sockeye	Naknek	Bering Sea	Comp	-0.05	0.05	0.15	5.24
sockeye	Nushagak	Bering Sea	Comp	-0.07	0.05	0.16	4.98
sockeye	Wood	Bering Sea	Comp	-0.02	0.07	0.17	7.07
sockeye	Alagnak	Bering Sea	Comp	-0.03	0.08	0.21	7.99
sockeye	Kvichak	Bering Sea	Comp	-0.10	0.02	0.12	2.26
sockeye	Togiak	Bering Sea	Comp	-0.04	0.05	0.14	5.07
sockeye	Goodnews	Bering Sea	Comp	-0.08	0.04	0.16	4.59
sockeye	Birkenhead	West Coast	SST x Comp	-0.13	-0.02	0.10	-1.63
sockeye	Bowron	West Coast	SST x Comp	-0.12	-0.01	0.12	-0.53
sockeye	Chilko	West Coast	SST x Comp	-0.09	0.01	0.13	0.59
sockeye	Cultus	West Coast	SST x Comp	-0.14	-0.01	0.12	-1.13
sockeye	Early Stuart	West Coast	SST x Comp	-0.07	0.03	0.17	2.94
sockeye	Gates	West Coast	SST x Comp	-0.16	-0.02	0.09	-2.45
sockeye	Harrison	West Coast	SST x Comp	-0.21	-0.04	0.07	-4.09
sockeye	Late Shuswap	West Coast	SST x Comp	-0.14	-0.02	0.10	-1.80
sockeye	Late Stuart	West Coast	SST x Comp	-0.11	0.00	0.14	-0.06
sockeye	Nadina	West Coast	SST x Comp	-0.10	0.01	0.15	0.64
sockeye	North Barriere	West Coast	SST x Comp	-0.10	0.00	0.13	-0.04
sockeye	Pitt	West Coast	SST x Comp	-0.18	-0.05	0.04	-4.58
sockeye	Portage	West Coast	SST x Comp	-0.12	-0.01	0.13	-0.58
sockeye	Quesnel	West Coast	SST x Comp	-0.14	-0.02	0.10	-1.75
sockeye	Raft	West Coast	SST x Comp	-0.15	-0.03	0.07	-2.71
sockeye	Scotch	West Coast	SST x Comp	-0.12	0.00	0.14	-0.27
sockeye	Seymour	West Coast	SST x Comp	-0.13	-0.01	0.11	-1.26
sockeye	Stellako	West Coast	SST x Comp	-0.11	0.00	0.11	-0.42
sockeye	Weaver	West Coast	SST x Comp	-0.12	0.00	0.14	-0.20

Table S3: Estimated stock-specific mean effect sizes for SST and Competitor covariates, and interaction term, from stationary models. *(continued)*

Species	Stock	Ocean Region	Coefficient	Lower CI	Mean	Upper CI	Mean % change in R/S
sockeye	Great Central Lake	West Coast	SST x Comp	-0.18	-0.04	0.07	-3.71
sockeye	Sproat Lake	West Coast	SST x Comp	-0.22	-0.05	0.06	-4.67
sockeye	Skidegate	West Coast	SST x Comp	-0.12	-0.01	0.09	-1.42
sockeye	Evelyn	West Coast	SST x Comp	-0.12	0.00	0.14	-0.13
sockeye	Awun	West Coast	SST x Comp	-0.20	-0.05	0.06	-4.47
sockeye	Yakoun	West Coast	SST x Comp	-0.13	-0.02	0.07	-2.27
sockeye	Alastair	Southeast Alaska	SST x Comp	-0.28	-0.09	0.01	-9.01
sockeye	Babine Early Wild	Southeast Alaska	SST x Comp	-0.15	-0.03	0.11	-2.91
sockeye	Babine Late Wild	Southeast Alaska	SST x Comp	-0.16	-0.04	0.09	-3.52
sockeye	Babine Mid Wild	Southeast Alaska	SST x Comp	-0.18	-0.05	0.06	-4.75
sockeye	Bear-Skeena	Southeast Alaska	SST x Comp	-0.13	-0.03	0.10	-2.74
sockeye	Kitsumkalum	Southeast Alaska	SST x Comp	-0.11	-0.02	0.08	-2.12
sockeye	Lakelse	Southeast Alaska	SST x Comp	-0.17	-0.05	0.05	-4.91
sockeye	Morice	Southeast Alaska	SST x Comp	-0.17	-0.05	0.06	-4.95
sockeye	Swan-Stephens	Southeast Alaska	SST x Comp	-0.16	-0.05	0.05	-4.73
sockeye	Bowser	Southeast Alaska	SST x Comp	-0.13	-0.02	0.10	-2.47
sockeye	Damdochax	Southeast Alaska	SST x Comp	-0.12	-0.03	0.08	-2.59
sockeye	Kwinageese	Southeast Alaska	SST x Comp	-0.16	-0.04	0.09	-3.63
sockeye	Lower Nass Sea & River Type	Southeast Alaska	SST x Comp	-0.21	-0.06	0.06	-5.71
sockeye	Meziadin	Southeast Alaska	SST x Comp	-0.17	-0.06	0.03	-5.35
sockeye	Upper Nass River Type	Southeast Alaska	SST x Comp	-0.22	-0.07	0.03	-6.74
sockeye	Redoubt	Southeast Alaska	SST x Comp	-0.12	-0.01	0.15	-1.17
sockeye	Chilkoot	Southeast Alaska	SST x Comp	-0.11	-0.01	0.14	-0.74
sockeye	Copper	Gulf of Alaska	SST x Comp	-0.12	-0.05	0.01	-4.48
sockeye	Early Russian	Gulf of Alaska	SST x Comp	-0.10	-0.03	0.06	-2.68
sockeye	Kenai	Gulf of Alaska	SST x Comp	-0.09	-0.03	0.05	-2.64
sockeye	Kasilof	Gulf of Alaska	SST x Comp	-0.10	-0.04	0.01	-3.93
sockeye	Frazer Lake	Gulf of Alaska	SST x Comp	-0.12	-0.03	0.05	-3.27
sockeye	Upper Station Early	Gulf of Alaska	SST x Comp	-0.10	-0.03	0.05	-2.92
sockeye	Upper Station Late	Gulf of Alaska	SST x Comp	-0.08	-0.02	0.07	-1.90
sockeye	Karluk Early	Gulf of Alaska	SST x Comp	-0.11	-0.04	0.03	-3.72
sockeye	Karluk Late	Gulf of Alaska	SST x Comp	-0.08	-0.02	0.06	-2.07
sockeye	Ayakulik	Gulf of Alaska	SST x Comp	-0.09	-0.03	0.05	-2.58
sockeye	Chignik Early	Gulf of Alaska	SST x Comp	-0.14	-0.05	0.02	-4.80
sockeye	Chignik Late	Gulf of Alaska	SST x Comp	-0.11	-0.04	0.02	-3.78
sockeye	Nelson River	Bering Sea	SST x Comp	-0.11	-0.02	0.05	-1.80
sockeye	Bear Lake	Bering Sea	SST x Comp	-0.07	0.00	0.09	0.23
sockeye	Ugashik	Bering Sea	SST x Comp	-0.10	-0.02	0.05	-1.69
sockeye	Egegik	Bering Sea	SST x Comp	-0.09	-0.01	0.06	-1.05
sockeye	Igushik	Bering Sea	SST x Comp	-0.07	0.00	0.07	-0.08
sockeye	Naknek	Bering Sea	SST x Comp	-0.08	-0.01	0.05	-1.11

Table S3: Estimated stock-specific mean effect sizes for SST and Competitor covariates, and interaction term, from stationary models. *(continued)*

Species	Stock	Ocean Region	Coefficient	Lower CI	Mean	Upper CI	Mean % change in R/S
sockeye	Nushagak	Bering Sea	SST x Comp	-0.06	0.00	0.08	0.19
sockeye	Wood	Bering Sea	SST x Comp	-0.06	0.00	0.06	-0.04
sockeye	Alagnak	Bering Sea	SST x Comp	-0.07	0.00	0.07	-0.23
sockeye	Kvichak	Bering Sea	SST x Comp	-0.09	-0.01	0.05	-1.34
sockeye	Togiak	Bering Sea	SST x Comp	-0.05	0.00	0.07	0.13
sockeye	Goodnews	Bering Sea	SST x Comp	-0.09	-0.01	0.06	-1.08
chum	Willapa Bay	West Coast	SST	-0.23	-0.10	0.07	-9.86
chum	Grays Harbour	West Coast	SST	-0.24	-0.10	0.08	-9.52
chum	S Sound Summer	West Coast	SST	-0.23	-0.10	0.09	-9.48
chum	S Sound Fall	West Coast	SST	-0.20	-0.08	0.09	-7.72
chum	S Sound Winter	West Coast	SST	-0.35	-0.17	-0.04	-15.79
chum	Hood Canal	West Coast	SST	-0.23	-0.09	0.08	-8.72
chum	Strait of Juan de Fuca	West Coast	SST	-0.53	-0.23	-0.07	-20.58
chum	Port Susan	West Coast	SST	-0.31	-0.14	0.00	-13.39
chum	Skagit	West Coast	SST	-0.31	-0.16	-0.02	-14.55
chum	Bellingham	West Coast	SST	-0.25	-0.12	0.00	-11.58
chum	Spiller-Fitz-Hugh-Burke	West Coast	SST	-0.22	-0.13	-0.04	-12.29
chum	Bella Coola-Dean Rivers	West Coast	SST	-0.17	-0.05	0.13	-4.44
chum	Mussel-Kynoch	West Coast	SST	-0.28	-0.16	-0.06	-14.90
chum	Hecate Lowlands	West Coast	SST	-0.26	-0.16	-0.06	-14.62
chum	Douglas-Gardner	West Coast	SST	-0.34	-0.18	-0.06	-16.24
chum	Afognak	Gulf of Alaska	SST	-0.24	0.03	0.28	3.35
chum	E Kodiak	Gulf of Alaska	SST	-0.12	0.03	0.19	3.50
chum	NW Kodiak	Gulf of Alaska	SST	-0.13	0.00	0.11	0.30
chum	Alitak	Gulf of Alaska	SST	-0.12	0.03	0.18	3.39
chum	SW Kodiak	Gulf of Alaska	SST	-0.25	0.00	0.19	-0.39
chum	Mainland Kodiak	Gulf of Alaska	SST	-0.09	0.04	0.19	4.01
chum	Chignik	Gulf of Alaska	SST	-0.07	0.06	0.23	5.69
chum	NW AK Peninsula	Bering Sea	SST	-0.06	0.03	0.13	3.35
chum	S AK Peninsula	Bering Sea	SST	0.00	0.09	0.20	9.89
chum	N AK Peninsula	Bering Sea	SST	-0.06	0.04	0.14	3.84
chum	Yukon - Summer	Bering Sea	SST	-0.21	-0.03	0.10	-2.80
chum	Yukon - Fall	Bering Sea	SST	-0.18	-0.01	0.11	-0.64
chum	Kwiniuk Inlet	Bering Sea	SST	-0.17	-0.01	0.11	-1.45
chum	Willapa Bay	West Coast	Comp	-0.21	-0.09	0.01	-8.69
chum	Grays Harbour	West Coast	Comp	-0.20	-0.08	0.03	-8.01
chum	S Sound Summer	West Coast	Comp	-0.22	-0.09	0.01	-8.74
chum	S Sound Fall	West Coast	Comp	-0.16	-0.07	0.06	-6.48
chum	S Sound Winter	West Coast	Comp	-0.24	-0.12	-0.02	-10.92
chum	Hood Canal	West Coast	Comp	-0.15	-0.06	0.11	-5.59

Table S3: Estimated stock-specific mean effect sizes for SST and Competitor covariates, and interaction term, from stationary models. (*continued*)

Species	Stock	Ocean Region	Coefficient	Lower CI	Mean	Upper CI	Mean % change in R/S
chum	Strait of Juan de Fuca	West Coast	Comp	-0.25	-0.12	0.01	-11.41
chum	Port Susan	West Coast	Comp	-0.22	-0.10	0.00	-9.73
chum	Skagit	West Coast	Comp	-0.27	-0.12	-0.02	-11.13
chum	Bellingham	West Coast	Comp	-0.17	-0.08	0.04	-7.79
chum	Spiller-Fitz-Hugh-Burke	West Coast	Comp	-0.18	-0.10	-0.02	-9.19
chum	Bella Coola-Dean Rivers	West Coast	Comp	-0.14	-0.05	0.10	-4.43
chum	Mussel-Kynoch	West Coast	Comp	-0.19	-0.10	-0.01	-9.66
chum	Hecate Lowlands	West Coast	Comp	-0.25	-0.12	-0.04	-11.51
chum	Douglas-Gardner	West Coast	Comp	-0.21	-0.11	0.01	-9.97
chum	Afognak	Gulf of Alaska	Comp	-0.11	0.07	0.38	7.42
chum	E Kodiak	Gulf of Alaska	Comp	-0.06	0.06	0.21	5.95
chum	NW Kodiak	Gulf of Alaska	Comp	-0.08	0.02	0.12	2.35
chum	Alitak	Gulf of Alaska	Comp	-0.05	0.07	0.23	6.84
chum	SW Kodiak	Gulf of Alaska	Comp	-0.24	0.00	0.16	0.46
chum	Mainland Kodiak	Gulf of Alaska	Comp	-0.09	0.04	0.16	4.09
chum	Chignik	Gulf of Alaska	Comp	-0.10	0.04	0.16	4.37
chum	NW AK Peninsula	Bering Sea	Comp	-0.15	-0.02	0.11	-1.81
chum	S AK Peninsula	Bering Sea	Comp	-0.04	0.10	0.25	10.28
chum	N AK Peninsula	Bering Sea	Comp	-0.16	-0.02	0.12	-1.76
chum	Yukon - Summer	Bering Sea	Comp	-0.34	-0.13	0.05	-11.90
chum	Yukon - Fall	Bering Sea	Comp	-0.24	-0.07	0.10	-6.86
chum	Kwiniuk Inlet	Bering Sea	Comp	-0.38	-0.13	0.07	-11.83
chum	Willapa Bay	West Coast	SST x Comp	-0.20	-0.10	-0.01	-9.37
chum	Grays Harbour	West Coast	SST x Comp	-0.19	-0.10	0.00	-9.10
chum	S Sound Summer	West Coast	SST x Comp	-0.19	-0.09	0.00	-8.93
chum	S Sound Fall	West Coast	SST x Comp	-0.19	-0.10	-0.01	-9.23
chum	S Sound Winter	West Coast	SST x Comp	-0.21	-0.10	-0.02	-9.64
chum	Hood Canal	West Coast	SST x Comp	-0.22	-0.10	-0.02	-9.94
chum	Strait of Juan de Fuca	West Coast	SST x Comp	-0.23	-0.11	-0.02	-10.06
chum	Port Susan	West Coast	SST x Comp	-0.21	-0.10	-0.02	-9.82
chum	Skagit	West Coast	SST x Comp	-0.17	-0.08	0.03	-7.88
chum	Bellingham	West Coast	SST x Comp	-0.25	-0.12	-0.04	-11.29
chum	Spiller-Fitz-Hugh-Burke	West Coast	SST x Comp	-0.15	-0.08	-0.01	-8.00
chum	Bella Coola-Dean Rivers	West Coast	SST x Comp	-0.16	-0.08	0.02	-7.78
chum	Mussel-Kynoch	West Coast	SST x Comp	-0.17	-0.09	-0.02	-8.74
chum	Hecate Lowlands	West Coast	SST x Comp	-0.16	-0.09	-0.01	-8.36
chum	Douglas-Gardner	West Coast	SST x Comp	-0.17	-0.09	0.00	-8.34
chum	Afognak	Gulf of Alaska	SST x Comp	-0.11	0.07	0.26	6.79

Table S3: Estimated stock-specific mean effect sizes for SST and Competitor covariates, and interaction term, from stationary models. (*continued*)

Species	Stock	Ocean Region	Coefficient	Lower CI	Mean	Upper CI	Mean % change in R/S
chum	E Kodiak	Gulf of Alaska	SST x Comp	-0.06	0.06	0.18	6.38
chum	NW Kodiak	Gulf of Alaska	SST x Comp	-0.02	0.07	0.16	6.92
chum	Alitak	Gulf of Alaska	SST x Comp	-0.04	0.07	0.20	7.55
chum	SW Kodiak	Gulf of Alaska	SST x Comp	-0.04	0.10	0.33	10.33
chum	Mainland Kodiak	Gulf of Alaska	SST x Comp	-0.05	0.06	0.16	6.07
chum	Chignik	Gulf of Alaska	SST x Comp	-0.10	0.03	0.13	3.41
chum	NW AK Peninsula	Bering Sea	SST x Comp	-0.13	-0.04	0.05	-4.37
chum	S AK Peninsula	Bering Sea	SST x Comp	-0.13	-0.05	0.04	-4.60
chum	N AK Peninsula	Bering Sea	SST x Comp	-0.18	-0.07	0.02	-6.93
chum	Yukon - Summer	Bering Sea	SST x Comp	-0.26	-0.09	0.02	-8.18
chum	Yukon - Fall	Bering Sea	SST x Comp	-0.17	-0.06	0.04	-6.08
chum	Kwiniuk Inlet	Bering Sea	SST x Comp	-0.19	-0.06	0.05	-6.10
pink	Nisqually-Odd	West Coast	SST	-0.28	-0.13	0.02	-12.49
pink	S South Misc.-Odd	West Coast	SST	-0.30	-0.15	-0.01	-13.73
pink	Puyallup-Odd	West Coast	SST	-0.27	-0.13	0.01	-12.24
pink	Hood Canal-Odd	West Coast	SST	-0.27	-0.13	0.02	-12.34
pink	Snohomish-Odd	West Coast	SST	-0.25	-0.11	0.04	-10.73
pink	Dungeness-Odd	West Coast	SST	-0.27	-0.13	0.02	-12.21
pink	Stillaguamish-Odd	West Coast	SST	-0.26	-0.12	0.02	-11.74
pink	Skagit-Odd	West Coast	SST	-0.26	-0.13	0.00	-12.31
pink	Nooksack-Odd	West Coast	SST	-0.27	-0.13	0.01	-12.37
pink	Fraser-Odd	West Coast	SST	-0.26	-0.12	0.05	-11.54
pink	Hecate	West Coast	SST	-0.32	-0.18	-0.05	-16.29
pink	Lowlands-Even						
pink	Hecate	West Coast	SST	-0.30	-0.16	-0.03	-15.17
pink	Strait-Fjords-Even						
pink	N Haida	Southeast Alaska	SST	-0.26	-0.02	0.20	-2.07
pink	Gwaii-Even						
pink	S Southeast Alaska-Even	Southeast Alaska	SST	-0.19	-0.02	0.13	-2.45
pink	S Southeast Alaska-Odd	Southeast Alaska	SST	-0.09	0.08	0.26	8.37
pink	N Southeast Alaska Outside-Even	Southeast Alaska	SST	-0.13	0.05	0.24	5.21
pink	N Southeast Alaska Outside-Odd	Southeast Alaska	SST	-0.08	0.11	0.34	11.70
pink	N Southeast Alaska Inside-Even	Southeast Alaska	SST	-0.18	0.01	0.21	0.51
pink	N Southeast Alaska Inside-Odd	Southeast Alaska	SST	-0.06	0.12	0.35	13.10
pink	Auke Creek-Even	Southeast Alaska	SST	-0.69	-0.26	0.08	-22.77
pink	Auke Creek-Odd	Southeast Alaska	SST	-0.67	-0.25	0.09	-21.74
pink	Prince William Sound-Even	Gulf of Alaska	SST	0.09	0.27	0.46	31.63

Table S3: Estimated stock-specific mean effect sizes for SST and Competitor covariates, and interaction term, from stationary models. (*continued*)

Species	Stock	Ocean Region	Coefficient	Lower CI	Mean	Upper CI	Mean % change in R/S
pink	Prince William Sound-Odd	Gulf of Alaska	SST	-0.02	0.17	0.35	18.68
pink	Kodiak-Even	Gulf of Alaska	SST	0.05	0.23	0.41	26.18
pink	Kodiak-Odd	Gulf of Alaska	SST	0.06	0.25	0.44	28.61
pink	Chignik-Even	Gulf of Alaska	SST	-0.23	0.02	0.28	2.33
pink	Chignik-Odd	Gulf of Alaska	SST	0.21	0.38	0.54	45.67
pink	S AKPen-Even	Gulf of Alaska	SST	0.07	0.30	0.56	35.18
pink	S AKPen-Odd	Gulf of Alaska	SST	0.33	0.59	0.85	81.07
pink	NW AK	Bering Sea	SST	0.10	0.49	0.93	64.00
	Peninsula-Even						
pink	NW AK	Bering Sea	SST	0.18	0.55	0.95	73.20
	Peninsula-Odd						
pink	Kwiniuk-Even	Bering Sea	SST	0.17	0.49	0.81	63.82
pink	Kwiniuk-Odd	Bering Sea	SST	0.12	0.53	0.98	69.17
pink	Nisqually-Odd	West Coast	Comp	-0.39	0.00	0.44	-0.36
pink	S South Misc.-Odd	West Coast	Comp	-0.42	-0.09	0.25	-8.90
pink	Puyallup-Odd	West Coast	Comp	-0.31	0.01	0.38	0.68
pink	Hood Canal-Odd	West Coast	Comp	-0.38	-0.04	0.32	-3.90
pink	Snohomish-Odd	West Coast	Comp	-0.20	0.13	0.50	13.73
pink	Dungeness-Odd	West Coast	Comp	-0.36	0.00	0.39	-0.39
pink	Stillaguamish-Odd	West Coast	Comp	-0.25	0.04	0.36	3.70
pink	Skagit-Odd	West Coast	Comp	-0.28	-0.01	0.27	-1.37
pink	Nooksack-Odd	West Coast	Comp	-0.33	-0.05	0.25	-4.65
pink	Fraser-Odd	West Coast	Comp	-0.32	-0.03	0.27	-2.89
pink	Hecate	West Coast	Comp	-0.60	-0.36	-0.09	-30.08
	Lowlands-Even						
pink	Hecate	West Coast	Comp	-0.62	-0.30	0.00	-25.81
	Strait-Fjords-Even						
pink	N Haida	Southeast Alaska	Comp	-0.04	0.16	0.32	17.00
	Gwaii-Even						
pink	S Southeast Alaska-Even	Southeast Alaska	Comp	0.01	0.16	0.29	16.94
pink	S Southeast Alaska-Odd	Southeast Alaska	Comp	0.09	0.23	0.42	26.06
pink	N Southeast Alaska Outside-Even	Southeast Alaska	Comp	0.05	0.21	0.39	22.93
pink	N Southeast Alaska Outside-Odd	Southeast Alaska	Comp	0.08	0.23	0.41	25.89
pink	N Southeast Alaska Inside-Even	Southeast Alaska	Comp	-0.02	0.16	0.31	17.46
pink	N Southeast Alaska Inside-Odd	Southeast Alaska	Comp	0.09	0.24	0.42	27.20
pink	Auke Creek-Even	Southeast Alaska	Comp	-0.26	0.04	0.26	4.41
pink	Auke Creek-Odd	Southeast Alaska	Comp	-0.24	0.05	0.26	5.18

Table S3: Estimated stock-specific mean effect sizes for SST and Competitor covariates, and interaction term, from stationary models. *(continued)*

Species	Stock	Ocean Region	Coefficient	Lower CI	Mean	Upper CI	Mean % change in R/S
pink	Prince William Sound-Even	Gulf of Alaska	Comp	-0.13	-0.01	0.12	-0.69
pink	Prince William Sound-Odd	Gulf of Alaska	Comp	-0.21	-0.06	0.07	-5.54
pink	Kodiak-Even	Gulf of Alaska	Comp	-0.18	-0.03	0.10	-3.06
pink	Kodiak-Odd	Gulf of Alaska	Comp	-0.14	-0.01	0.13	-0.91
pink	Chignik-Even	Gulf of Alaska	Comp	-0.38	-0.14	0.04	-13.30
pink	Chignik-Odd	Gulf of Alaska	Comp	-0.08	0.04	0.17	3.81
pink	S AKPen-Even	Gulf of Alaska	Comp	-0.17	0.00	0.16	-0.45
pink	S AKPen-Odd	Gulf of Alaska	Comp	-0.04	0.16	0.42	17.43
pink	NW AK Peninsula-Even	Bering Sea	Comp	-0.48	0.06	0.50	6.25
pink	NW AK Peninsula-Odd	Bering Sea	Comp	-0.20	0.22	0.73	24.12
pink	Kwiniuk-Even	Bering Sea	Comp	-0.23	0.14	0.54	15.28
pink	Kwiniuk-Odd	Bering Sea	Comp	-0.39	0.10	0.55	10.70
pink	Nisqually-Odd	West Coast	SST x Comp	-0.50	-0.20	0.05	-18.47
pink	S South Misc.-Odd	West Coast	SST x Comp	-0.58	-0.25	-0.02	-22.28
pink	Puyallup-Odd	West Coast	SST x Comp	-0.40	-0.17	0.07	-15.40
pink	Hood Canal-Odd	West Coast	SST x Comp	-0.35	-0.12	0.20	-10.97
pink	Snohomish-Odd	West Coast	SST x Comp	-0.45	-0.21	0.00	-18.74
pink	Dungeness-Odd	West Coast	SST x Comp	-0.49	-0.20	0.05	-18.34
pink	Stillaguamish-Odd	West Coast	SST x Comp	-0.52	-0.24	-0.03	-21.46
pink	Skagit-Odd	West Coast	SST x Comp	-0.51	-0.24	-0.04	-21.40
pink	Nooksack-Odd	West Coast	SST x Comp	-0.46	-0.21	0.00	-19.25
pink	Fraser-Odd	West Coast	SST x Comp	-0.29	-0.07	0.20	-7.19
pink	Hecate Lowlands-Even	West Coast	SST x Comp	-0.31	-0.14	0.04	-13.39
pink	Hecate Strait-Fjords-Even	West Coast	SST x Comp	-0.38	-0.16	0.07	-15.03
pink	N Haida Gwaii-Even	Southeast Alaska	SST x Comp	-0.29	-0.14	0.01	-13.09
pink	S Southeast Alaska-Even	Southeast Alaska	SST x Comp	-0.26	-0.14	-0.01	-13.11
pink	S Southeast Alaska-Odd	Southeast Alaska	SST x Comp	-0.29	-0.15	-0.03	-14.27
pink	N Southeast Alaska Outside-Even	Southeast Alaska	SST x Comp	-0.24	-0.12	0.04	-11.16
pink	N Southeast Alaska Outside-Odd	Southeast Alaska	SST x Comp	-0.28	-0.14	0.00	-13.05
pink	N Southeast Alaska Inside-Even	Southeast Alaska	SST x Comp	-0.30	-0.15	-0.01	-13.83
pink	N Southeast Alaska Inside-Odd	Southeast Alaska	SST x Comp	-0.30	-0.15	-0.02	-14.16
pink	Auke Creek-Even	Southeast Alaska	SST x Comp	-0.30	-0.13	0.06	-12.41

Table S3: Estimated stock-specific mean effect sizes for SST and Competitor covariates, and interaction term, from stationary models. (*continued*)

Species	Stock	Ocean Region	Coefficient	Lower CI	Mean	Upper CI	Mean % change in R/S
pink	Auke Creek-Odd	Southeast Alaska	SST x Comp	-0.36	-0.16	-0.01	-14.80
pink	Prince William Sound-Even	Gulf of Alaska	SST x Comp	-0.36	-0.15	0.05	-14.27
pink	Prince William Sound-Odd	Gulf of Alaska	SST x Comp	-0.11	0.09	0.28	9.20
pink	Kodiak-Even	Gulf of Alaska	SST x Comp	-0.21	0.01	0.26	0.95
pink	Kodiak-Odd	Gulf of Alaska	SST x Comp	-0.29	-0.10	0.09	-9.92
pink	Chignik-Even	Gulf of Alaska	SST x Comp	-0.26	-0.01	0.27	-0.74
pink	Chignik-Odd	Gulf of Alaska	SST x Comp	-0.38	-0.21	-0.04	-18.90
pink	S AKPen-Even	Gulf of Alaska	SST x Comp	-0.48	-0.23	0.01	-20.32
pink	S AKPen-Odd	Gulf of Alaska	SST x Comp	-0.46	-0.24	-0.03	-21.53
pink	NW AK	Bering Sea	SST x Comp	-0.58	-0.17	0.31	-15.39
	Peninsula-Even						
pink	NW AK	Bering Sea	SST x Comp	-0.72	-0.27	0.13	-23.82
	Peninsula-Odd						
pink	Kwiniuk-Even	Bering Sea	SST x Comp	-0.54	-0.27	0.00	-23.52
pink	Kwiniuk-Odd	Bering Sea	SST x Comp	-0.52	-0.03	0.71	-2.70

Table S4: Estimated stock-specific mean effect sizes for SST and Competitor covariates, from time-varying ‘era’ models.

Species	Ocean Region	Coefficient	Era	Lower CI	Mean	Upper CI
sockeye	West Coast	SST	Early	-0.10	0.00	0.10
sockeye	Southeast Alaska	SST	Early	-0.08	0.06	0.20
sockeye	Gulf of Alaska	SST	Early	-0.01	0.10	0.20
sockeye	Bering Sea	SST	Early	0.08	0.19	0.30
sockeye	West Coast	SST	Middle	-0.22	-0.13	-0.04
sockeye	Southeast Alaska	SST	Middle	-0.05	0.05	0.16
sockeye	Gulf of Alaska	SST	Middle	-0.06	0.05	0.15
sockeye	Bering Sea	SST	Middle	0.16	0.27	0.39
sockeye	West Coast	SST	Late	-0.47	-0.31	-0.15
sockeye	Southeast Alaska	SST	Late	-0.44	-0.29	-0.14
sockeye	Gulf of Alaska	SST	Late	-0.28	-0.14	-0.01
sockeye	Bering Sea	SST	Late	0.05	0.18	0.30
sockeye	West Coast	Competitors	Stationary	-0.35	-0.28	-0.20
sockeye	Southeast Alaska	Competitors	Stationary	-0.11	-0.03	0.05
sockeye	Gulf of Alaska	Competitors	Stationary	-0.09	0.00	0.08
sockeye	Bering Sea	Competitors	Stationary	-0.02	0.06	0.14
chum	West Coast	SST	Early	-0.15	-0.05	0.04
chum	Gulf of Alaska	SST	Early	-0.42	-0.14	0.11
chum	Bering Sea	SST	Early	-0.02	0.12	0.24
chum	West Coast	SST	Middle	-0.27	-0.18	-0.10
chum	Gulf of Alaska	SST	Middle	-0.20	0.00	0.17

Table S4: Estimated stock-specific mean effect sizes for SST and Competitor covariates, from time-varying ‘era’ models. (*continued*)

Species	Ocean Region	Coefficient	Era	Lower CI	Mean	Upper CI
chum	Bering Sea	SST	Middle	-0.09	0.06	0.21
chum	West Coast	SST	Late	-0.46	-0.32	-0.17
chum	Gulf of Alaska	SST	Late	-0.09	0.09	0.29
chum	Bering Sea	SST	Late	-0.23	-0.09	0.04
chum	West Coast	Competitors	Stationary	-0.16	-0.08	-0.01
chum	Gulf of Alaska	Competitors	Stationary	-0.04	0.07	0.19
chum	Bering Sea	Competitors	Stationary	-0.26	-0.06	0.10
pink	West Coast	SST	Early	-0.27	-0.03	0.20
pink	Southeast Alaska	SST	Early	0.03	0.18	0.33
pink	Gulf of Alaska	SST	Early	0.18	0.41	0.63
pink	Bering Sea	SST	Early	0.45	1.09	1.70
pink	West Coast	SST	Middle	-0.49	-0.29	-0.11
pink	Southeast Alaska	SST	Middle	-0.15	0.02	0.19
pink	Gulf of Alaska	SST	Middle	-0.05	0.20	0.45
pink	Bering Sea	SST	Middle	-0.20	0.38	0.99
pink	West Coast	SST	Late	-0.64	-0.38	-0.11
pink	Southeast Alaska	SST	Late	-0.38	-0.21	-0.05
pink	Gulf of Alaska	SST	Late	-0.08	0.17	0.40
pink	Bering Sea	SST	Late	-0.25	0.27	0.84
pink	West Coast	Competitors	Stationary	-0.22	0.00	0.22
pink	Southeast Alaska	Competitors	Stationary	0.02	0.17	0.32
pink	Gulf of Alaska	Competitors	Stationary	-0.19	-0.03	0.13
pink	Bering Sea	Competitors	Stationary	-0.57	0.00	0.56

Table S5. Estimated stock-specific mean effect sizes for SST and Competitor covariates, from time-varying ‘era’ models.

- need to figure out what to present, not worth doing all years is it?

Table S6. Estimated regional mean effect sizes for SST and Competitor covariates, from annually time-varying (random walk) models. - need to figure out what to present, not worth doing all years is it?

Table S8. Priors on all directly estimated parameters for each model type.

See: *Output/Model_priors.xlsx*

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